Question - 13: Write a Python program to implement the Minimax algorithm for gaming.

Program Code

```
    def minimax(depth, node, maximizing, values,
max_depth):
        if depth == max_depth:
            return values[node]

        if maximizing:
            best = float("-inf")

            for i in range(2):
                value = minimax(
                    depth + 1,
                    node * 2 + i,
                    False,
                    values,
                    max_depth
                )
                best = max(best, value)

            return best

        else:
            best = float("inf")

            for i in range(2):
                value = minimax(
                    depth + 1,
                    node * 2 + i,
                    True,
```

```
            values,
            max_depth
        )
        best = min(best, value)

    return best


leaf_values = [3, 5, 2, 9, 12, 5, 23, 23]

result = minimax(0, 0, True, leaf_values, 3)

print("Leaf Values:", leaf_values)
print("Best Value using Minimax:", result)
```

Program Output

```
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB> &
ython/Python314/python.exe "e:/Desktop/SIMATS/Artifici
max.py"
Leaf Values: [3, 5, 2, 9, 12, 5, 23, 23]
Best Value using Minimax: 12
PS E:\Desktop\SIMATS\Artificial Intelligence\AI LAB>
```